

RCV Ballot Measure Models

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```
library(dplyr)
library(ggplot2)
library(lme4)
library(scales)
library(broom)
library(broom.mixed)
library(knitr)

load("../data/rcv_data.RData")
```

Data Preparation

The cleaning pipeline (R/01_clean_and_merge.R) now produces unified `rcv_yes` and `rcv_no` columns across all 8 locales.

```
df <- states_and_cities %>%
  filter(!is.na(rcv_yes), !is.na(rcv_no), !is.na(dem_share),
         (rcv_yes + rcv_no) > 0) %>%
  mutate(
    n_votes = rcv_yes + rcv_no,
    recent_lpw = factor(recent_lpw, levels = c(FALSE, TRUE),
                        labels = c("No", "Yes"))
  )

df %>%
  group_by(locale, rcv_jurisdiction, recent_lpw) %>%
  summarise(n_precincts = n(), total_votes = sum(n_votes),
            mean_yes = mean(yes_share), .groups = "drop") %>%
  kable(digits = 3)
```

locale	rcv_jurisdiction	recent_lpw	n_precincts	total_votes	mean_yes
Alaska	statewide	Yes	578	344283	0.486
Albany	local	No	3	9765	0.733
Bloomington	local	No	32	49497	0.512
Boulder	local	No	88	54756	0.782
Eureka	local	No	8	11602	0.603
Maine	statewide	Yes	527	739201	0.456
Massachusetts	statewide	No	2173	3427366	0.470
Minnetonka	local	No	23	33768	0.547

1 – Binomial GLM (pooled)

A binomial GLM with a logit link uses the actual vote counts as the response, properly accounting for the bounded $[0, 1]$ nature of `yes_share` and giving more weight to precincts with more voters.

```
m_binom <- glm(
  cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw,
  family = binomial(link = "logit"),
  data = df
)

summary(m_binom)
```

Call:

```
glm(formula = cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw,
     family = binomial(link = "logit"), data = df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-2.506041	0.005041	-497.1	<2e-16 ***
dem_share	3.552460	0.007444	477.2	<2e-16 ***
recent_lpwYes	0.932374	0.002717	343.2	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 312814 on 3431 degrees of freedom
Residual deviance: 59898 on 3429 degrees of freedom
AIC: 84453

Number of Fisher Scoring iterations: 4

The deviance-to-df ratio measures overdispersion ($\gg 1$ means the binomial variance assumption is too tight – common with aggregate vote data).

```
cat("Residual deviance / df:",
    round(deviance(m_binom) / df.residual(m_binom), 1), "\n")
```

Residual deviance / df: 17.5

Overdispersion is expected: precinct vote shares vary more than pure binomial sampling because voters within a precinct are not independent coin flips. A **quasibinomial** model uses the same point estimates but corrects the standard errors.

```
m_quasi <- glm(
  cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw,
  family = quasibinomial(link = "logit"),
  data = df
)

tidy(m_quasi, conf.int = TRUE) %>%
  kable(digits = 4, caption = "Quasibinomial GLM: yes ~ dem_share + recent_lpw")
```

Table 2: Quasibinomial GLM: $\text{yes} \sim \text{dem_share} + \text{recent_lpw}$

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2.5060	0.0210	-119.0606	0	-2.5473	-2.4648
dem_share	3.5525	0.0311	114.2959	0	3.4916	3.6134
recent_lpwYes	0.9324	0.0113	82.2028	0	0.9102	0.9546

Adding the interaction (dem_share x recent_lpw)

Does the effect of partisanship on RCV support differ in locales with a recent low-plurality winner?

```
m_quasi_int <- glm(
  cbind(rcv_yes, rcv_no) ~ dem_share * recent_lpw,
  family = quasibinomial(link = "logit"),
  data = df
)

tidy(m_quasi_int, conf.int = TRUE) %>%
  kable(digits = 4,
    caption = "Quasibinomial GLM with interaction")
```

Table 3: Quasibinomial GLM with interaction

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-2.7646	0.0238	-116.0576	0	-2.8113	-2.7180
dem_share	3.9434	0.0354	111.3304	0	3.8741	4.0129
recent_lpwYes	1.6120	0.0349	46.1431	0	1.5435	1.6804
dem_share:recent_lpwYes	-1.3063	0.0637	-20.5219	0	-1.4310	-1.1815

```
anova(m_quasi, m_quasi_int, test = "F")
```

Analysis of Deviance Table

Model 1: $\text{cbind}(\text{rcv_yes}, \text{rcv_no}) \sim \text{dem_share} + \text{recent_lpw}$

Model 2: $\text{cbind}(\text{rcv_yes}, \text{rcv_no}) \sim \text{dem_share} * \text{recent_lpw}$

	Resid. Df	Resid. Dev	Df	Deviance	F	Pr(>F)
1	3429	59898				
2	3428	53368	1	6529.8	415.29	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Marginal effects plot

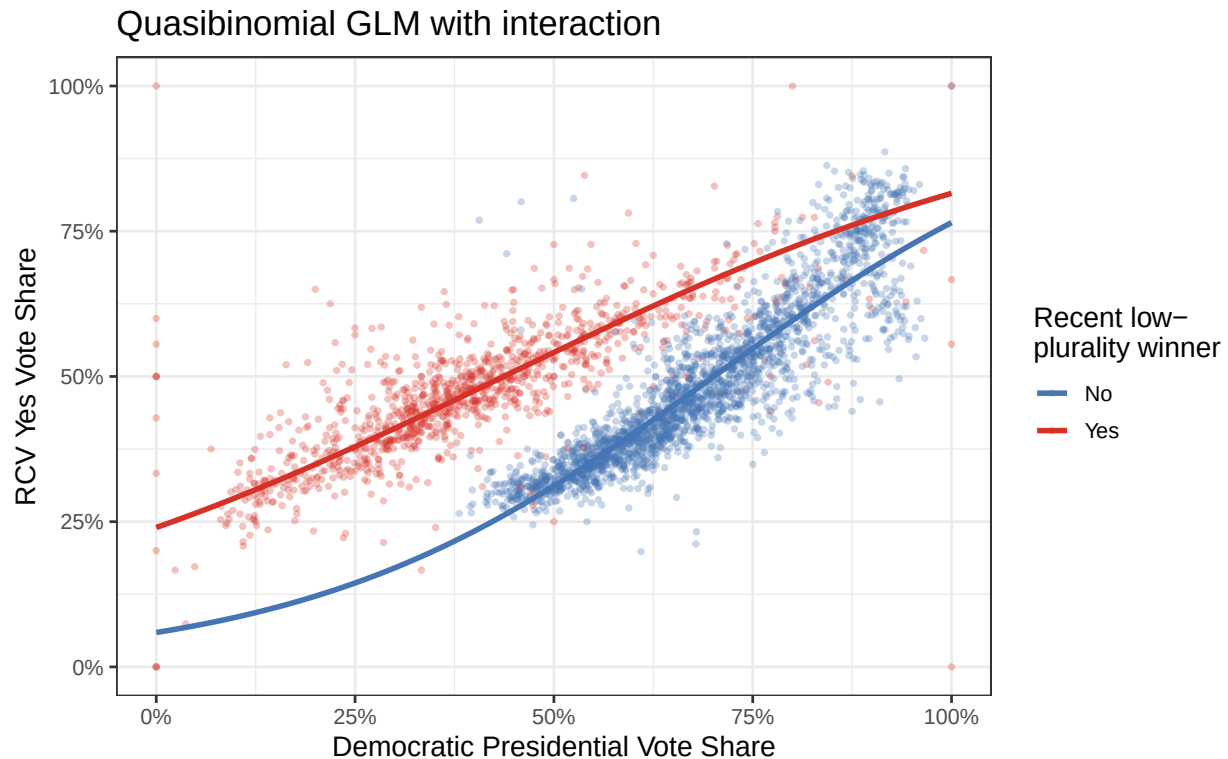
```
pred_grid <- expand.grid(
  dem_share = seq(0, 1, length.out = 200),
  recent_lpw = factor(c("No", "Yes"), levels = c("No", "Yes"))
)
```

```

pred_grid$fit <- predict(m_quasi_int, newdata = pred_grid, type = "response")

ggplot() +
  geom_point(data = df,
    aes(x = dem_share, y = yes_share, color = recent_lpw),
    alpha = 0.3, size = 0.8) +
  geom_line(data = pred_grid,
    aes(x = dem_share, y = fit, color = recent_lpw),
    linewidth = 1.2) +
  scale_x_continuous(labels = percent_format(1)) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  scale_color_manual(values = c("No" = "#4575B4", "Yes" = "#D73027")) +
  labs(x = "Democratic Presidential Vote Share",
    y = "RCV Yes Vote Share",
    color = "Recent low-\nplurality winner",
    title = "Quasibinomial GLM with interaction") +
  theme_bw(base_size = 13)

```



2 – Mixed-effects binomial model (random intercepts by locale)

Precincts are nested within locales that differ in unobserved ways (campaign spending, ballot language, media environment). A random-intercept model accounts for this clustering.

```

m_glmr <- glmer(
  cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw + (1 | locale),
  family = binomial,
  data = df,

```

```
nAGQ = 10
)

summary(m_glmer)
```

```
Generalized linear mixed model fit by maximum likelihood (Adaptive
Gauss-Hermite Quadrature, nAGQ = 10) [glmerMod]
Family: binomial ( logit )
Formula: cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw + (1 | locale)
Data: df
```

```
      AIC      BIC    logLik -2*log(L)  df.resid
51468.2   51492.7  -25730.1   51460.2    3428
```

Scaled residuals:

```
      Min      1Q  Median      3Q      Max
-16.777  -1.859   0.102   2.004  44.041
```

Random effects:

```
Groups Name      Variance Std.Dev.
locale (Intercept) 0.03554  0.1885
Number of obs: 3432, groups: locale, 8
```

Fixed effects:

```
              Estimate Std. Error z value Pr(>|z|)
(Intercept)  -2.093009   0.077384 -27.047  < 2e-16 ***
dem_share      3.463857   0.007591 456.322  < 2e-16 ***
recent_lpwYes  0.578503   0.154067   3.755 0.000173 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Correlation of Fixed Effects:

```
      (Intr) dm_shr
dem_share  -0.072
recnt_lpwYs -0.501  0.014
```

```
tidy(m_glmer, conf.int = TRUE, effects = "fixed") %>%
  kable(digits = 4, caption = "GLMM: random intercept by locale")
```

Table 4: GLMM: random intercept by locale

effect	term	estimate	std.error	statistic	p.value	conf.low	conf.high
fixed	(Intercept)	-2.0930	0.0774	-27.0472	0e+00	-2.2447	-1.9413
fixed	dem_share	3.4639	0.0076	456.3217	0e+00	3.4490	3.4787
fixed	recent_lpwYes	0.5785	0.1541	3.7549	2e-04	0.2765	0.8805

Random-intercept estimates (locale-level deviations on the log-odds scale):

```
ranef(m_glmer)$locale %>%
  tibble::rownames_to_column("locale") %>%
  rename(intercept_shift = `(Intercept)`) %>%
```

```
arrange(desc(intercept_shift)) %>%
kable(digits = 3)
```

locale	intercept_shift
Boulder	0.332
Eureka	0.150
Alaska	0.048
Minnetonka	-0.013
Albany	-0.020
Maine	-0.048
Bloomington	-0.075
Massachusetts	-0.374

With interaction

```
m_glmer_int <- glmer(
  cbind(rcv_yes, rcv_no) ~ dem_share * recent_lpw + (1 | locale),
  family = binomial,
  data = df,
  nAGQ = 10
)

tidy(m_glmer_int, conf.int = TRUE, effects = "fixed") %>%
  kable(digits = 4,
        caption = "GLMM with interaction + random intercept")
```

Table 6: GLMM with interaction + random intercept

effect	term	estimate	std.error	statistic	p.value	conf.low	conf.high
fixed	(Intercept)	-2.3553	0.0710	-33.1845	0	-2.4944	-2.2162
fixed	dem_share	3.8195	0.0092	417.0989	0	3.8016	3.8375
fixed	recent_lpwYes	1.2022	0.1413	8.5102	0	0.9253	1.4791
fixed	dem_share:recent_lpwYes	-1.1566	0.0163	-71.0371	0	-1.1885	-1.1247

```
anova(m_glmer, m_glmer_int)
```

Data: df

Models:

m_glmer: cbind(rcv_yes, rcv_no) ~ dem_share + recent_lpw + (1 | locale)

m_glmer_int: cbind(rcv_yes, rcv_no) ~ dem_share * recent_lpw + (1 | locale)

	npar	AIC	BIC	logLik	-2*log(L)	Chisq	Df	Pr(>Chisq)
m_glmer	4	51468	51493	-25730	51460			
m_glmer_int	5	46481	46512	-23236	46471	4989	1	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Random slopes

Does the effect of `dem_share` vary by locale? Adding a random slope lets each locale have its own partisan gradient.

```
m_glmer_rs <- glmer(
  cbind(rcv_yes, rcv_no) ~ dem_share * recent_lpw + (1 + dem_share | locale),
  family = binomial,
  data = df,
  nAGQ = 1 # Laplace; nAGQ > 1 not supported with random slopes
)

summary(m_glmer_rs)
```

```
Generalized linear mixed model fit by maximum likelihood (Laplace
Approximation) [glmerMod]
Family: binomial ( logit )
Formula: cbind(rcv_yes, rcv_no) ~ dem_share * recent_lpw + (1 + dem_share |
  locale)
Data: df
```

AIC	BIC	logLik	-2*log(L)	df.resid
70902.7	70945.6	-35444.3	70888.7	3425

Scaled residuals:

Min	1Q	Median	3Q	Max
-16.892	-1.768	-0.054	1.690	46.435

Random effects:

Groups Name	Variance	Std.Dev.	Corr
locale (Intercept)	1.877	1.370	
dem_share	2.261	1.504	-0.99

Number of obs: 3432, groups: locale, 8

Fixed effects:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.35320	0.62728	-2.157	0.030987 *
dem_share	2.68382	0.69147	3.881	0.000104 ***
recent_lpwYes	0.20292	1.13723	0.178	0.858386
dem_share:recent_lpwYes	-0.02053	1.24972	-0.016	0.986890

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Correlation of Fixed Effects:

	(Intr)	dm_shr	rcnt_Y
dem_share	-0.991		
recent_lpwYes	-0.541	0.536	
dm_shr:rc_Y	0.537	-0.542	-0.990

```
coef(m_glmer_rs)$locale %>%
  tibble::rownames_to_column("locale") %>%
  kable(digits = 4, caption = "Locale-specific coefficients (random slopes)")
```

Table 7: Locale-specific coefficients (random slopes)

locale	(Intercept)	dem_share	recent_lpwYes	dem_share:recent_lpwYes
Alaska	-1.3840	2.8186	0.2029	-0.0205
Albany	1.3778	-0.4089	0.2029	-0.0205
Bloomington	-2.3914	3.8136	0.2029	-0.0205
Boulder	-0.0097	1.4634	0.2029	-0.0205
Eureka	-2.5111	4.2863	0.2029	-0.0205
Maine	-1.3224	2.5490	0.2029	-0.0205
Massachusetts	-2.7028	3.8219	0.2029	-0.0205
Minnetonka	-1.8827	3.1267	0.2029	-0.0205

Fitted curves from the random-slopes model

```

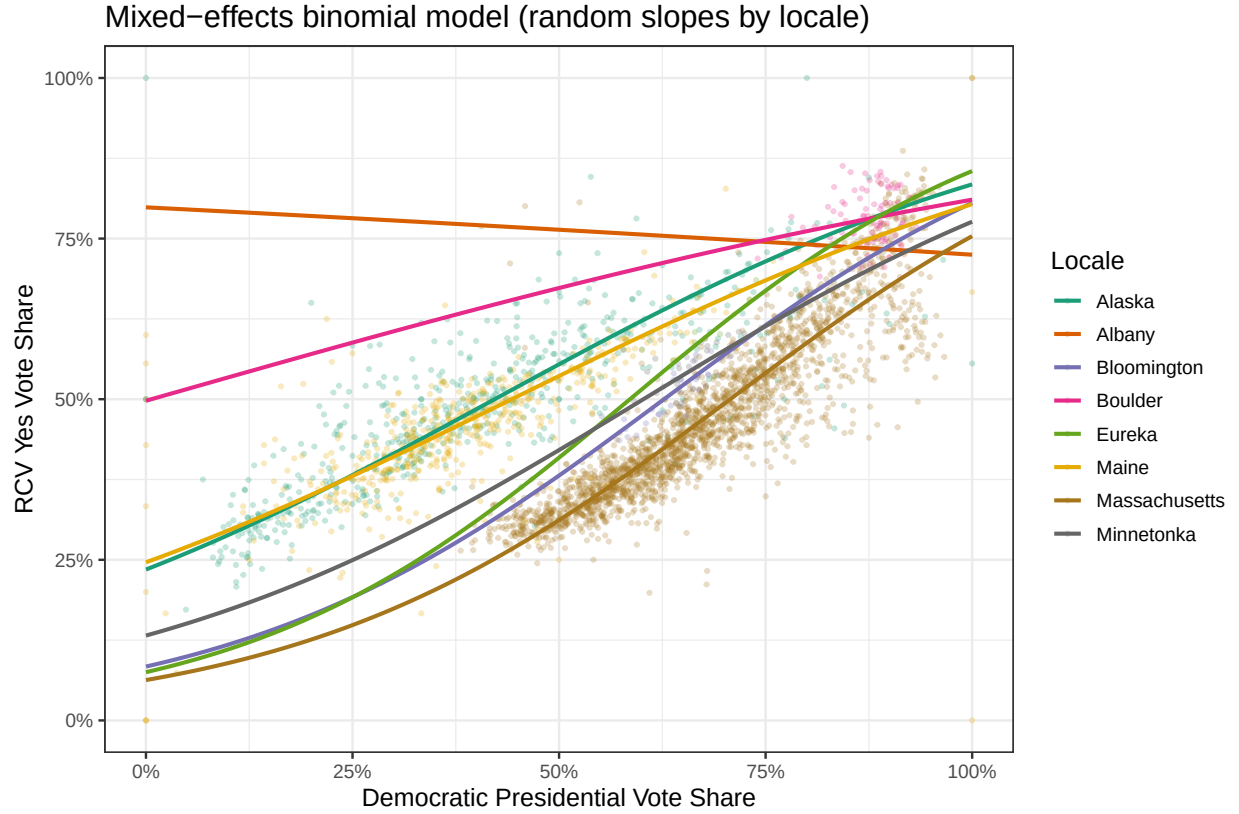
locale_coefs <- coef(m_glmmer_rs)$locale %>%
  tibble::rownames_to_column("locale")

locale_meta <- df %>%
  distinct(locale, recent_lpw)

pred_rs <- locale_coefs %>%
  left_join(locale_meta, by = "locale") %>%
  tidyr::crossing(dem_share_grid = seq(0, 1, length.out = 200)) %>%
  mutate(
    lpw_val = ifelse(recent_lpw == "Yes", 1, 0),
    eta = `(Intercept)` + dem_share * dem_share_grid +
      `recent_lpwYes` * lpw_val +
      `dem_share:recent_lpwYes` * dem_share_grid * lpw_val,
    fitted = plogis(eta)
  )

ggplot() +
  geom_point(data = df,
    aes(x = dem_share, y = yes_share, color = locale),
    alpha = 0.25, size = 0.7) +
  geom_line(data = pred_rs,
    aes(x = dem_share_grid, y = fitted, color = locale),
    linewidth = 1) +
  scale_x_continuous(labels = percent_format(1)) +
  scale_y_continuous(labels = percent_format(1), limits = c(0, 1)) +
  scale_color_brewer(palette = "Dark2") +
  labs(x = "Democratic Presidential Vote Share",
    y = "RCV Yes Vote Share",
    color = "Locale",
    title = "Mixed-effects binomial model (random slopes by locale)") +
  theme_bw(base_size = 13)

```

3 – Model comparison summary

AIC is undefined for quasibinomial models, so the comparison below includes only the GLMM fits (which have proper likelihoods). The quasibinomial models are compared via the F-test above.

```
models <- tibble(
  Model = c(
    "GLMM random intercept",
    "GLMM random intercept + interaction",
    "GLMM random slopes + interaction"
  ),
  AIC = c(AIC(m_glmr), AIC(m_glmr_int), AIC(m_glmr_rs))
)

models %>%
  mutate(AIC = format(round(AIC), big.mark = ",")) %>%
  kable(caption = "AIC comparison (lower is better)")
```

Table 8: AIC comparison (lower is better)

Model	AIC
GLMM random intercept	51,468
GLMM random intercept + interaction	46,481
GLMM random slopes + interaction	70,903